FanWise AI: A Software-Based Intelligent Fan Optimization Platform

atomQUEST 2026 · Track 1: Fan · Round 1 Solution Proposal

Team The Catalyst · VIT Bhopal University

A. Team & Institutional Context

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B. Problem & Idea Framing

1. Problem Statement

Atomberg's BLDC fans are efficient, but blind: they can be told what to do, yet cannot sense whether they should run, or at what speed. Users adjust speed manually, often faster or longer than needed, with no visibility into what a setting costs.

2. Proposed Solution

FanWise AI is a software platform that helps users understand and improve fan usage. Users enter fan specifications, operating speed, daily usage, tariff and room conditions through a web interface. It provides:

- Consumption and cost estimation from the supplied inputs.
- A rule-based engine recommending suitable settings from user preferences, occupancy and entered conditions.
- A dashboard showing estimated consumption, projected costs and scenario comparisons.
- A scenario simulator showing how duration and speed changes affect estimated consumption.

Built around Atomberg's public developer platform. Enabled through Developer Mode in the Atomberg Home app, it reads the user's real device state and usage analytics, so recommendations are anchored to the fan actually owned. The quota of **100 API calls per day** shapes the design: reads are batched and cached locally. The first release runs on user inputs and labelled simulated data — no sensors, extra hardware or trained model.

3. Target Performance

Parameter	Target
Calculation reliability	100% agreement with independently calculated results across valid energy-consumption test cases.
Recommendation response	A recommendation generated within 2 seconds under normal local testing conditions.
Scenario comparison	Compare at least three fan-usage scenarios.
Usability	Enter specifications, view estimated costs and understand recommendations.
Energy-saving analysis	Projected energy and cost differences between user-defined scenarios.

All figures are targets until verified by testing; estimated savings will not be presented as experimentally measured results.

C. Technical Feasibility

FanWise AI is a web application: HTML and CSS for the interface, Python and Flask for calculation and recommendation logic. The backend processes inputs, performs energy and cost calculations, compares scenarios and returns results to the interface.

Energy (kWh/day) = Power (W) × Hours per day ÷ 1000 | Cost = Energy × Tariff (per kWh)

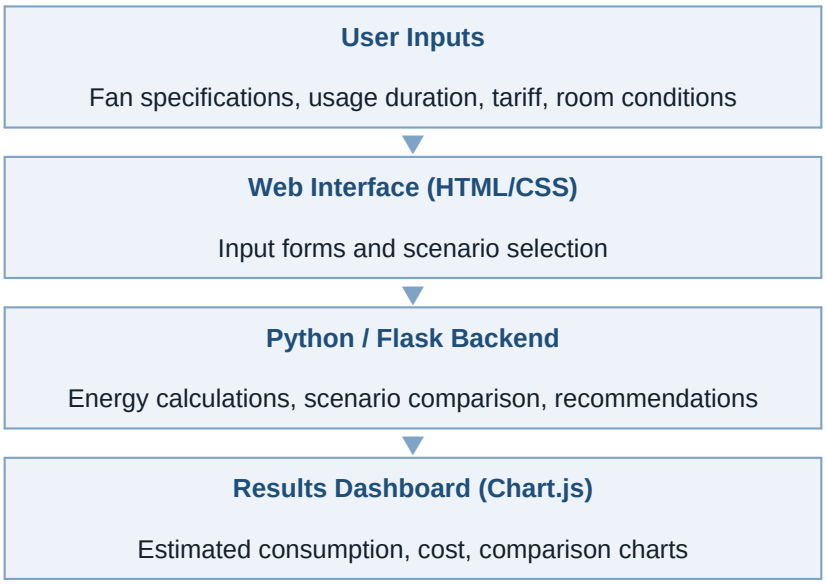
Where reliable speed-specific power data is unavailable, assumptions are explicitly labelled, not treated as measured. Chart.js dashboards display estimated consumption, cost and scenario comparisons; SQLite optionally stores saved scenarios. The initial version uses predefined rules, not machine learning, so no dataset is needed. Testing uses known examples, boundary cases and simulated scenarios.

D. Execution Feasibility

1. Software Components

Component	Purpose
HTML and CSS	Web interface and page styling
Python	Energy calculations and recommendation logic
Flask	Backend application and routing
Chart.js	Dashboard visualizations
CSV or JSON	Sample fan specifications and simulated scenarios
SQLite (optional)	Storage of fan specifications and saved scenarios
Git and GitHub	Version control and project collaboration

2. System-Level Block Diagram



3. Data Requirements

No machine-learning dataset is required. The application runs on user-supplied specifications and labelled sample scenarios; public technical specifications are documented at source. Measured power data for specific models would improve speed-wise accuracy later.

4. Cost and Resource Requirements

The prototype uses free tools only — a code editor, Python, Flask and browser testing. No sensors, microcontrollers or physical components are needed, so development cost is minimal.

5. Implementation Plan for Round 2

Phase	Activity
1	Build the input interface; implement energy and cost calculations.
2	Integrate the recommendation engine for user-defined scenarios.
3	Develop the interactive dashboard with comparison charts.
4	Validate against independently computed examples and boundary cases; document model limitations.
5	Package as a working web app with screenshots, diagram, test results and demo instructions.

E. Expected Outcome

FanWise AI demonstrates how software can help users understand fan energy consumption and make more informed decisions, with an accessible interface for comparing scenarios, estimating costs and exploring savings. The first release is a standalone demonstrator, with potential future integration with compatible smart fans or sensors.